Bias and Hallucination Risks Analysis

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CEN4090L: Software Engineering Capstone Project

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1. Overview

Artificial Medical Intelligence (A.M.I.) is a conversational AI assistant designed to provide educational information on personalized medicine while maintaining transparency through peer-reviewed citations and ethical safeguards. Because A.M.I. relies on large language models (LLMs) and multiple medical datasets, there are inherent risks of bias, hallucinations, misinformation, and unsafe outputs. Hallucinations occur when an AI generates responses that appear correct but are unsupported or incorrect, while bias occurs when the AI reflects imbalances or limitations in its training data.

2. Bias Risks

Dataset Bias

- MedQuAD focuses primarily on general consumer health information, which may overrepresent common conditions and underrepresent rare diseases.
- PubMedQA uses peer-reviewed abstracts, but abstract-level information may lack full clinical detail.
- HealthCareMagic-100k and MedDialog contain conversational doctor-patient responses without verified citations, which may introduce subjective or outdated information.
- MedAESQA provides peer-reviewed citations but has limited size, reducing coverage.

Population Bias

- Medical datasets often overrepresent Western populations and adult demographics, which may reduce accuracy for underrepresented populations.

Model Bias

- Language models learn patterns from training data and may produce generalized or skewed responses if training data is unbalanced.

Interface Bias

- The friendly virtual avatar may increase user trust, potentially causing users to treat responses as medical advice rather than educational information.

3. Hallucination Risks

Generative Model Hallucination

- Language models generate responses probabilistically, which can result in fabricated or incorrect medical information.

Conversational Dataset Risk

- Datasets without citations may cause unsupported claims in generated responses.

Prompt Ambiguity

- Incomplete or vague user prompts may cause the AI to make incorrect assumptions.

Scope Violations

- If users request diagnosis or treatment advice, hallucination risk increases.

Model Accuracy Limitations

- Smaller models may have reduced accuracy and increased hallucination probability.

4. Privacy and Safety Risks

Although datasets are de-identified, patterns may still reflect sensitive scenarios. The system must avoid storing personally identifiable information unless securely encrypted.

5. Risk Mitigation Strategies

• Use Retrieval-Augmented Generation (RAG) with peer-reviewed sources

• Always provide citations for responses

• Filter and prioritize reliable datasets

• Implement prompt guardrails

• Provide medical disclaimers

• Collect user feedback

• Conduct validation and testing

6. Residual Risk Assessment

Dataset bias: Medium risk

Hallucination risk: Medium risk after mitigation

User over-trust risk: Medium risk

Prompt misuse risk: Low risk after mitigation

7. Conclusion

Bias and hallucination risks cannot be fully eliminated but can be significantly reduced through careful dataset selection, citation transparency, system testing, and guardrails. These measures ensure A.M.I. provides reliable and ethical educational medical information.